Summary
Details
Source
Context
Comments
Raw
Session
Compare
Tools
View
Export
GPU Throughput
Compute (SM) [%]
Memory [%]
0.0
10.0
20.0
30.0
40.0
50.0
60.0
70.0
80.0
90.0
100.0
Speed Of Light (SOL) [%]
Floating Point Operations Roofline
Floating Point Operations Roofline (Double Precision)
Floating Point Operations Roofline (Half Precision)
Floating Point Operations Roofline (Single Precision)
HW Performance [FLOP/s] (1 = 1E+12)
HW Arithmetic Intensity [FLOP/byte]
100
10
1
0.1
0.01
1,000
10,000
Overview
SM Frequency
Total Warps Launched
Warps Active
Total Blocks Launched
Blocks Active
Total CGAs Launched
CGAs Active
SM
Executed IPC
SM Throughput
SM Throughput by Pipeline
SM Shared Bandwidth
SM DCC Hit Miss
Warp Stall Reasons (All Samples)
L1
L1 Writeback Active
L1 Lookup Throughput
L1 Lookup Hit Miss
L1 Throughput (Data)
L1 Throughput (Data) by Access
SMEM Bank Conflicts
L2
L2 Throughput
L2 Sectors Throughput
L2 Sectors Throughput by Source
L2 to XBAR Active
XBAR to L2 Active
L2 Hit Miss
L2 Hit Rate by Source Unit
SysL2
SysL2 Throughput
SysL2 Sectors Throughput
SysL2 Sectors Throughput by Source
SysL2 Atomic Input Active
SysL2 Hit Miss
SysL2 to XBAR Active
XBAR to SysL2 Active
DRAM
DRAM Throughput
DRAM Bandwidth
DRAM Sectors Bandwidth
Interconnect
PCIe Throughput
PCIe Bandwidth
Workload Execution
2.65 Ghz
1 warp/cycle
4.03k cycle
0.07 block
504 block
1.42 inst/cycle
100 %
16.3 Gbyte/s
1.46k issue
32
100 sector
946 Gbyte/s
29.6 sector/ns
10 Gbyte/s
0s
+0.2ms
+0.4ms
+0.6ms
+0.8ms
+1ms
+1.2ms
+1.4ms
+1.6ms
+1.8ms
+2ms
+2.2ms
sparseCE_v3_online
Pipe Utilization (% of elapsed cycles)
Pipe Utilization (% of peak instructions executed over elapsed cycles)
Pipe Utilization (% of active cycles)
Pipe Utilization (% of peak instructions executed over active cycles)
FMA
ALU
FP64
TMA
Tensor (All)
Tensor (FP)
Tensor (INT)
XU
LSU
FMA (FP16)
ADU
CBU
Uniform
FP64 (DMMA)
FP64 (FP64)
TEX
WorkID/CLC
0.0
25.0
50.0
75.0
100.0
Utilization [%]
Memory Chart
Values: Throughput
Inactivity: Hidden
Kernel
Global
Shared
DSMEM
12.90 M Inst
12.89 M Req
8.19 K Req
262.14 K Inst
114.69 K Req
147.46 K Req
461.15 K Req
401.44 K Req
90.11 K Req
L1/TEX Cache Hit Rate: 0.00 %
Shared Memory
ICC Hit Rate: 97.74 %
DCC Hit Rate: 99.12 %
IDC Hit Rate: 98.22 %
877.71 GB/s
139.49 MB/s
De-compressor Rate: 1.00
Compressor Rate: 0.00
Device L2 Cache
882.08 GB/s
3.95 GB/s
Device Memory
66.30 KB
2.80 K Req
336.00 Req
GCC Hit Rate: 82.68 %
% Peak
100%
80%
60%
40%
20%
0%

Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current 581 - sparseCE_v3_online	(8192, 1, 1)x(256, 1, 1)	1.88 ms	49,51,808	0 - NVIDIA GeForce RTX 5080	2.63 Ghz	[53078] sparseCE	

GPU Speed Of Light Throughput

All

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	26.35	Duration [ms]	1.88
Memory Throughput [%]	93.63	Elapsed Cycles [cycle]	49,51,808
L1/TEX Cache Throughput [%]	12.73	SM Active Cycles [cycle]	48,19,461.24
L2 Cache Throughput [%]	36.25	SM Frequency [Ghz]	2.63
DRAM Throughput [%]	93.63	DRAM Frequency [Ghz]	14.79

High Throughput This workload is utilizing greater than 80.0% of the available compute or memory performance of this device. To further improve performance, work will likely need to be shifted from the most utilized to another unit. Start by analyzing DRAM in the ▸Memory Workload Analysis section.

Roofline Analysis The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved 10% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Profiling Guide for more details on roofline analysis.

Compute Throughput Breakdown

SM: Inst Executed [%]	26.35
SM: Issue Active [%]	26.35
SM: Pipe Fmaheavy Cycles Active [%]	21.45
SM: Pipe Fma Cycles Active [%]	16.03
SM: Pipe Aluheavy Cycles Active [%]	12.77
SM: Inst Executed Pipe Xu [%]	12.77
SM: Inst Executed Pipe Lsu [%]	6.71
SM: Mio Inst Issued [%]	2.32
SM: Mio2rf Writeback Active [%]	0.84
SM: Inst Executed Pipe Adu [%]	0.24
SM: Mio Pq Write Cycles Active [%]	0.20
SM: Mio Pq Read Cycles Active [%]	0.20
SM: Inst Executed Pipe Cbu Pred On Any [%]	0.16
SM: Memory Throughput Internal Activity [%]	0.10
IDC: Request Cycles Active [%]	0.04
SM: Inst Executed Pipe Uniform [%]	0.03
SM: Instruction Throughput Internal Activity [%]	0
SM: Inst Executed Pipe Tex [%]	0
SM: Inst Executed Pipe Ipa [%]	0
SM: Pipe Fp64 Cycles Active [%]	0
SM: Pipe Tensor Cycles Active [%]	0
SM: Pipe Tma Cycles Active [%]	0

Memory Throughput Breakdown

DRAM: Cycles Active [%]	93.63
DRAM: Dram Sectors [%]	73.20
L2: D Sectors Fill Device [%]	36.25
L2: D Sectors [%]	18.17
L2: T Sectors [%]	18.13
L2: Lts2xbar Cycles Active [%]	18.13
L1: M Xbar2l1tex Read Sectors [%]	12.40
GPU: Compute Memory Access Throughput Internal Activity [%]	9.08
L2: T Tag Requests [%]	9.08
L2: Xbar2lts Cycles Active [%]	8.85
L1: Lsuin Requests [%]	6.71
L1: Data Pipe Lsu Wavefronts [%]	3.37
L1: Lsu Writeback Active [%]	3.30
L1: M L1tex2xbar Req Cycles Active [%]	2.84
L1: Data Bank Writes [%]	1.55
L1: Data Bank Reads [%]	1.55
GPU: Compute Memory Request Throughput Internal Activity [%]	0.01
SYSLTS: Xbar2syslts Cycles Active [%]	0.01
SYSLTS: Syslts2xbar Cycles Active [%]	0.01
SYSLTS: D Sectors Fill Sysmem [%]	0.01
SYSLTS: T Sectors [%]	0.01
SYSLTS: D Sectors [%]	0.00
SYSLTS: T Tag Requests [%]	0.00
L1: Texin Sm2tex Req Cycles Active [%]	0.00
L2: D Decomp Input Sectors [%]	0
L2: D Atomic Input Cycles Active [%]	0
L1: Tmain Requests [%]	0
L1: Tex Writeback Active [%]	0
L1: M Xbar2l1tex Read Sectors Mem Dshared [%]	0
L1: F Wavefronts [%]	0
SYSLTS: D Atomic Input Cycles Active [%]	0
L1: Data Pipe Tex Wavefronts [%]	0

Tensor Core Operations Roofline

All roofline charts in this section body are disabled.

	# Operations	# Operations / Cycle	# Operations / s	Peak %	Peak Operations / Cycle	Peak Operations / s
Op:hmma Src:bf16 Dst:fp32 Sparsity:on	0	0	0	0	86,016	22,65,92,05,12,38,37…
Op:hmma Src:fp16 Dst:fp32 Sparsity:off	0	0	0	0	86,016	22,65,92,05,12,38,37…
Op:hmma Src:fp16 Dst:fp16 Sparsity:on	0	0	0	0	1,72,032	45,31,84,10,24,76,74…
Op:hmma Src:fp16 Dst:fp16 Sparsity:off	0	0	0	0	43,008	11,32,96,02,56,19,18…
Op:hmma Src:fp16 Dst:fp32 Sparsity:on	0	0	0	0	86,016	22,65,92,05,12,38,37…
Op:hmma Src:tf32 Dst:fp32 Sparsity:off	0	0	0	0	21,504	5,66,48,01,28,09,593.56
Op:hmma Src:tf32 Dst:fp32 Sparsity:on	0	0	0	0	43,008	11,32,96,02,56,19,18…
Op:imma Src:int8 Sparsity:off	0	0	0	0	1,72,032	45,31,84,10,24,76,74…
Op:imma Src:int8 Sparsity:on	0	0	0	0	3,44,064	90,63,68,20,49,53,497
Src:bf16 Dst:fp32	0	0	0	0	86,016	22,65,92,05,12,38,37…
Src:fp16 Dst:fp16	0	0	0	0	1,72,032	45,31,84,10,24,76,74…
Src:fp16 Dst:fp32	0	0	0	0	86,016	22,65,92,05,12,38,37…
Src:fp4 Dst:fp32 Sparsity:off	0	0	0	0	3,44,064	90,63,68,20,49,53,497
Src:fp4 Dst:fp32 Sparsity:on	0	0	0	0	6,88,128	1,81,27,36,40,99,06,994
Src:fp4,fp6 Dst:fp16 Sparsity:off	0	0	0	0	86,016	22,65,92,05,12,38,37…
Src:fp4,fp6 Dst:fp16 Sparsity:on	0	0	0	0	1,72,032	45,31,84,10,24,76,74…
Src:fp4,fp6 Dst:fp32 Sparsity:off	0	0	0	0	86,016	22,65,92,05,12,38,37…
Src:fp4,fp6 Dst:fp32 Sparsity:on	0	0	0	0	1,72,032	45,31,84,10,24,76,74…
Src:fp64	0	0	0	0	294.58	7,76,00,01,75,473.88
Src:fp8 Dst:fp16 Sparsity:off	0	0	0	0	1,72,032	45,31,84,10,24,76,74…
Src:fp8 Dst:fp16 Sparsity:on	0	0	0	0	3,44,064	90,63,68,20,49,53,497
Src:fp8 Dst:fp32 Sparsity:off	0	0	0	0	86,016	22,65,92,05,12,38,37…
Src:fp8 Dst:fp32 Sparsity:on	0	0	0	0	1,72,032	45,31,84,10,24,76,74…
Src:int8	0	0	0	0	3,44,064	90,63,68,20,49,53,497
Src:tf32 Dst:fp32	0	0	0	0	43,008	11,32,96,02,56,19,18…

PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [us]	3	# Pass Groups	2
Maximum Buffer Size [Mbyte]	33.36	-	

Compute Workload Analysis

All

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	1.05	SM Busy [%]	26.35
Executed Ipc Active [inst/cycle]	1.08	Issue Slots Busy [%]	26.35
Issued Ipc Active [inst/cycle]	1.08		

Low Utilization
Est. Local Speedup: 83.97% All compute pipelines are under-utilized. Either this workload is very small or it doesn't issue enough warps per scheduler. Check the ▸Launch Statistics and ▸Scheduler Statistics sections for further details.

▸ Key Performance Indicators

Memory Workload Analysis

Memory Chart

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [Gbyte/s]	886.03	Mem Busy [%]	18.17
L1/TEX Hit Rate [%]	0.00	Max Bandwidth [%]	93.63
L2 Hit Rate [%]	0.05	Mem Pipes Busy [%]	6.71
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Shared Memory Spilling Requests	0
L2 Compression Success Rate [%]	0	Local Memory Spilling Request Overhead [%]	0
L2 Persisting Size [Mbyte]	12.58	Shared Memory Spilling Request Overhead [%]	0
L2 Sector Promotion Misses [%]	0	-	

DRAM Global Load Access Pattern
Est. Speedup: 0.03% The memory access pattern for global loads from DRAM might not be optimal. On average, only 32.0 of the 32 bytes transmitted per sector are utilized by each thread. This applies to the 100.0% of sectors missed in L2. This could possibly be caused by a stride between threads. Check the ▸Source Counters section for uncoalesced global loads.

▸ Key Performance Indicators

L2 Global Store Access Pattern
Est. Speedup: 25.98% The memory access pattern for global stores to L2 might not be optimal. On average, only 4.0 of the 32 bytes transmitted per sector are utilized by each thread. This applies to the 81.9% of sectors missed in L1TEX. This could possibly be caused by a stride between threads. Check the ▸Source Counters section for uncoalesced global stores.

▸ Key Performance Indicators

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	11.88	No Eligible [%]	73.00
Eligible Warps Per Scheduler [warp]	0.37	One or More Eligible [%]	27.00
Issued Warp Per Scheduler	0.27		

Issue Slot Utilization
Est. Local Speedup: 6.37% Every scheduler is capable of issuing one instruction per cycle, but for this workload each scheduler only issues an instruction every 3.7 cycles. This might leave hardware resources underutilized and may lead to less optimal performance. Out of the maximum of 12 warps per scheduler, this workload allocates an average of 11.88 active warps per scheduler, but only an average of 0.37 warps were eligible per cycle. Eligible warps are the subset of active warps that are ready to issue their next instruction. Every cycle with no eligible warp results in no instruction being issued and the issue slot remains unused. To increase the number of eligible warps, avoid possible load imbalances due to highly different execution durations per warp. Reducing stalls indicated on the ▸Warp State Statistics and ▸Source Counters sections can help, too.

▸ Key Performance Indicators

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle]	43.99	Avg. Active Threads Per Warp	31.81
Warp Cycles Per Executed Instruction [cycle]	43.99	Avg. Not Predicated Off Threads Per Warp	31.77

Long Scoreboard Stalls
Est. Speedup: 6.37% On average, each warp of this workload spends 40.6 cycles being stalled waiting for a scoreboard dependency on a L1TEX (local, global, surface, texture) operation. Find the instruction producing the data being waited upon to identify the culprit. To reduce the number of cycles waiting on L1TEX data accesses verify the memory access patterns are optimal for the target architecture, attempt to increase cache hit rates by increasing data locality (coalescing), or by changing the cache configuration. Consider moving frequently used data to shared memory. This stall type represents about 92.2% of the total average of 44.0 cycles between issuing two instructions.

▸ Key Performance Indicators

Warp Stall Check the ▸Warp Stall Sampling (All Samples) table for the top stall locations in your source based on sampling data. The Profiling Guide provides more details on each stall reason.

Instruction Statistics

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	43,81,73,696	Avg. Executed Instructions Per Scheduler [inst]	13,04,088.38
Issued Instructions [inst]	43,81,73,696	Avg. Issued Instructions Per Scheduler [inst]	13,04,088.38
Local Memory Spilling Requests [inst]	0	Shared Memory Spilling Requests [inst]	0

FP32 Non-Fused Instructions
Est. Speedup: 3.21% This workload executes 119463936 fused and 80003072 non-fused FP32 instructions. By converting pairs of non-fused instructions to their fused, higher-throughput equivalent, the achieved FP32 performance could be increased by up to 20% (relative to its current performance).

▾ Most frequently executed non-fused FP32 instructions

Location	Opcode	Executed Instructions
sparseCE.cu, line 176	FADD	5.15113E+07
sparseCE.cu, line 176	FMUL	2.57556E+07
sparseCE.cu, line 187	FADD	655360

▸ Key Performance Indicators

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Tables

Detailed tables with properties for each NVLink.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	8,192	Function Cache Configuration	CachePreferNone
Cluster Size	0	Preferred Cluster Size	0
Registers Per Thread [register/thread]	33	Cluster Scheduling Policy	PolicySpread
Static Shared Memory Per Block [byte/block]	72	Block Size	256
Dynamic Shared Memory Per Block [byte/block]	0	Threads [thread]	20,97,152
Driver Shared Memory Per Block [kbyte/block]	1.02	Waves Per SM	16.25
Shared Memory Configuration Size [Kbyte]	16.38	Uses Green Context	0
Stack Size	1,024	# SMs [SM]	84
# TPCs	42	Enabled TPC IDs	all

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	6
Theoretical Active Warps per SM [warp]	48	Block Limit Shared Mem [block]	14
Achieved Occupancy [%]	100.18	Block Limit Warps [block]	6
Achieved Active Warps Per SM [warp]	48.09	Block Limit SM [block]	24
Cluster Occupancy [%]	0	Block Limit Barriers [block]	24
Max Active Clusters [cluster]	0	Max Cluster Size [block]	8
Overall GPU Occupancy [%]	0		

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	48,19,461.24	Average L1 Active Cycles [cycle]	48,19,461.24
Average L2 Active Cycles [cycle]	43,81,551.09	Average SMSP Active Cycles [cycle]	48,30,022.42
Average DRAM Active Cycles [cycle]	2,60,07,608	Total SM Elapsed Cycles [cycle]	41,56,97,280
Total L1 Elapsed Cycles [cycle]	41,56,97,280	Total L2 Elapsed Cycles [cycle]	14,21,26,976
Total SMSP Elapsed Cycles [cycle]	1,66,27,89,120	Total DRAM Elapsed Cycles [cycle]	22,22,26,432

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	48,19,461.24	47,27,210	48,91,460	40,48,34,744
SMSP Active Cycles	48,30,022.42	47,43,252	48,95,678	1,62,28,87,532
L1 Active Cycles	48,19,461.24	47,27,210	48,91,460	40,48,34,744
L2 Active Cycles	43,81,551.09	43,79,738	43,83,804	14,02,09,635
DRAM Active Cycles	2,60,07,608	2,60,05,856	2,60,09,632	20,80,60,864

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	1,35,98,720	Branch Efficiency [%]	99.44
Branch Instructions Ratio [%]	0.03	Avg. Divergent Branches [branches]	219.43

Warp Stall Sampling (Not-issued Samples)

Location	Value	Value (%)
sparseCE.cu:175 (0x7116cb76cf60 in sparseCE_v3...	1,35,413	95
sparseCE.cu:195 (0x7116cb76d950 in sparseCE_v3...	1,288	1
sparseCE.cu:176 (0x7116cb76d0a0 in sparseCE_v3...	949	1
sparseCE.cu:174 (0x7116cb76cee0 in sparseCE_v3...	509	0
sparseCE.cu:174 (0x7116cb76cf20 in sparseCE_v3...	406	0

Most Instructions Executed

Location	Value	Value (%)
sparseCE.cu:173 (0x7116cb76d0e0 in sparseCE_v3...	1,28,77,824	0
sparseCE.cu:176 (0x7116cb76d0d0 in sparseCE_v3...	1,28,77,824	0
sparseCE.cu:173 (0x7116cb76d0c0 in sparseCE_v3...	1,28,77,824	0
sparseCE.cu:176 (0x7116cb76d0b0 in sparseCE_v3...	1,28,77,824	0
sparseCE.cu:176 (0x7116cb76d0a0 in sparseCE_v3...	1,28,77,824	0

Follow the *rules outputs* to get guidance on how to navigate through the report and quickly discover performance bottlenecks in this kernel.
You could also disable individual sections to focus on selected performance aspects and make profiling faster.

You can ask Insight Copilot questions about this report.